

Product LONDON

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1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : LONDON
IFF Code : 5965487
SDS Number : 300000814541

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Fragrance for consumer product

1.3 Details of the supplier of the safety data sheet

Company : IFF (NEDERLAND) BV
ZEVENHEUVELENWEG 60
5048 AN TILBURG
Netherlands
Telephone : +31134642211
Telefax : +31134636032
E-mail address : sds@iff.com
Responsible/issuing person

1.4 Emergency telephone number

Refer to section 16 for country specific emergency contact number.

2. Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Skin irritation, Category 2	H315: Causes skin irritation.
Eye irritation, Category 2	H319: Causes serious eye irritation.
Skin sensitisation, Category 1	H317: May cause an allergic skin reaction.
Reproductive toxicity, Category 2	H361: Suspected of damaging fertility or the unborn child.
Short-term (acute) aquatic hazard, Category 1	H400: Very toxic to aquatic life.
Long-term (chronic) aquatic hazard, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

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Hazard pictograms

:



Signal word

: Warning

Hazard statements

: H315 H317 H319 H361 H410	Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation. Suspected of damaging fertility or the unborn child. Very toxic to aquatic life with long lasting effects.
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Precautionary statements

: Prevention: P201 P261 P273 P280 Response: P308 + P313 P362 + P364	Obtain special instructions before use. Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray. Avoid release to the environment. Wear protective gloves/ protective clothing/ eye protection/ face protection. IF exposed or concerned: Get medical advice/ attention. Take off contaminated clothing and wash it before reuse.
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Hazardous components which must be listed on the label:

- 68155-66-8, 54464-57-2, 68155-67-9, 54464-59-4 Tetramethyl Acetyloctahydronaphthalenes
- 32388-55-9 cedr-8-enyl methyl ketone
- 33704-61-9 Dihydro pentamethylindanone
- 67874-81-1 [3R-(3 α ,3 β ,6 β ,7 β ,8 $\alpha\alpha$)]-octahydro-6-methoxy-3,6,8,8-tetramethyl-1H-3a,7-methanoazulene
- 1205-17-0 α -methyl-1,3-benzodioxole-5-propionaldehyde
- 138-86-3, 5989-27- limonene

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- 5, 5989-54-8
- 91-64-5 COUMARIN
- 476332-65-7 Heptamethyl Decahydroindenofuran
- 80-56-8, 7785-26-4 pin-2(3)-ene
- 115-95-7 linalyl acetate
- 127-91-3 pin-2(10)-ene
- 78-70-6, 126-91-0 Linalool
- 87-44-5 Beta-Caryophyllene
- 4707-47-5 methyl 2,4-dihydroxy-3,6-dimethylbenzoate
- 13466-78-9 3,7,7-trimethylbicyclo[4.1.0]hept-3-ene
- 104-55-2 CINNAMAL
- 23696-85-7 1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one
- 97-54-1 ISOEUGENOL

2.3 Other hazards

None reasonably foreseeable.

3. Composition/information on ingredients

3.1 Substances

Not applicable, product is a mixture.

3.2 Mixtures

Hazardous components

Chemical name	CAS-No. EC-No. Registration number	Classification (REGULATION (EC) No 1272/2008)	Concentration [%]
Tetramethyl Acetyloctahydronaphthalenes	68155-66-8, 54464-57- 2, 68155-67-9, 54464- 59-4 915-730-3 01-2119489989-04	Skin Irrit.2; H315 Skin Sens.1B; H317 Aquatic Chronic1; H410	20 - 25
cedr-8-enyl methyl ketone	32388-55-9 251-020-3 01-2119969651-28	Skin Sens.1B; H317 Aquatic Acute1; H400 Aquatic Chronic1; H410	10 - 20
Dihydro pentamethylindanone	33704-61-9 251-649-3 01-2119977131-40	Aquatic Chronic2; H411 Skin Sens.1B; H317 Eye Irrit.2; H319	5 - 10

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		Skin Irrit.2; H315	
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran	1222-05-5 214-946-9 01-2119488227-29	Aquatic Chronic1; H410 Aquatic Acute1; H400	2,5 - 10
[3R-(3 α ,3 α β ,6 β ,7 β ,8 $\alpha\alpha$)]-octahydro-6-methoxy-3,6,8,8-tetramethyl-1H-3a,7-methanoazulene	67874-81-1 267-510-5 01-2120228335-61	Skin Sens.1B; H317 Aquatic Acute1; H400 Aquatic Chronic1; H410	2,5 - 10
2,6-dimethyloct-7-en-2-ol	18479-58-8 242-362-4 01-2119457274-37	Eye Irrit.2; H319 Skin Irrit.2; H315	5 - 10
2-ethyl-4-(2,2,3-trimethyl-3-cyclopenten-1-yl)-2-buten-1-ol	28219-61-6 701-122-3 01-2119529224-45	Eye Irrit.2; H319 Aquatic Chronic2; H411	3 - 10
α -methyl-1,3-benzodioxole-5-propionaldehyde	1205-17-0 214-881-6	Skin Sens.1B; H317 Aquatic Chronic2; H411 Repr.2; H361	3 - 10
limonene	138-86-3, 5989-27-5, 5989-54-8 205-341-0, 227-813-5, 227-815-6	Flam. Liq.3; H226 Skin Sens.1B; H317 Asp. Tox.1; H304 Aquatic Chronic1; H410 Skin Irrit.2; H315 Aquatic Acute1; H400	2,5 - 5
coumarin	91-64-5 202-086-7 01-2119943756-26	Acute Tox.4; H302 Skin Sens.1B; H317	1 - 10
vanillin	121-33-5 204-465-2 01-2119516040-60	Eye Irrit.2; H319	1 - 3
4h-Indeno [4,5-d]-1,3-dioxole, 3a,5,6,7,8,8b-hexahydro- 2,2,6,6,7,8,8-heptamethyl	823178-41-2 457-070-4, 457-070-4 01-0000019329-63	Aquatic Chronic2; H411	1 - 2,5
3-methyl-4-(2,6,6-trimethyl-2-cyclohexen-1-yl)-3-buten-2-one	127-51-5 204-846-3	Aquatic Chronic2; H411	1 - 2,5

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decahydro heptamethyl indenofuran	476332-65-7 449-360-4 01-0000018977-51	Skin Sens.1B; H317 Aquatic Chronic1; H410	1 - 2,5
pin-2(3)-ene	80-56-8, 7785-26-4 201-291-9, 232-077-3 01-2119519223-49	Flam. Liq.3; H226 Asp. Tox.1; H304 Skin Irrit.2; H315 Skin Sens.1B; H317 Aquatic Acute1; H400 Aquatic Chronic1; H410 Acute Tox.4; H302	0,25 - 1
linalyl acetate	115-95-7 204-116-4	Skin Irrit.2; H315 Skin Sens.1; H317 Eye Irrit.2; H319	0,1 - 1
pin-2(10)-ene	127-91-3 242-060-2, 204-872-5 01-2119519230-54	Asp. Tox.1; H304 Skin Irrit.2; H315 Flam. Liq.3; H226 Skin Sens.1B; H317 Aquatic Acute1; H400 Aquatic Chronic1; H410	0,25 - 1
linalool	78-70-6, 126-91-0 201-134-4 01-2119474016-42	Skin Irrit.2; H315 Skin Sens.1B; H317 Eye Irrit.2; H319	0,1 - 1
caryophyllene	87-44-5 201-746-1	Asp. Tox.1; H304 Skin Sens.1B; H317 Aquatic Chronic4; H413	0,25 - 1
7-methyl-3-methyleneocta- 1,6-diene	123-35-3 204-622-5 01-2119514321-56	Asp. Tox.1; H304 Skin Irrit.2; H315 Flam. Liq.3; H226 Eye Irrit.2; H319 Aquatic Chronic2; H411 Aquatic Acute1; H400	0,1 - 0,25
methyl 2,4-dihydroxy-3,6- dimethylbenzoate	4707-47-5 225-193-0 01-2120762759-36	Skin Sens.1B; H317	0,1 - 1
3,7,7- trimethylbicyclo[4.1.0]hept-	13466-78-9 236-719-3	Flam. Liq.3; H226 Asp. Tox.1; H304	0,1 - 0,25

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3-ene		Skin Sens.1; H317 Skin Irrit.2; H315 Aquatic Chronic1; H410 Aquatic Acute1; H400	
cinnamaldehyde	104-55-2 203-213-9 01-2119935242-45-	Acute Tox.4; H312 Skin Irrit.2; H315 Eye Irrit.2; H319 Skin Sens.1A; H317	0,1 - 1
1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one	23696-85-7 245-833-2	Skin Sens.1A; H317 Aquatic Chronic2; H411 Skin Irrit.2; H315	0,025 - 0,1
isoeugenol	97-54-1 202-590-7	Acute Tox.4; H312 Acute Tox.4; H302 Skin Irrit.2; H315 Eye Irrit.2; H319 Skin Sens.1A; H317 Acute Tox.4; H332 STOT SE3; H335	0 - 0,01

For the full text of the R-phrases mentioned in this Section, see Section 16.

Components with workplace control parameters

Components	EC-No.	Concentration [%]
limonene, dipentene, d-limonene, (R)-p-mentha-1,8-diene	205-341-0, 227-813-5, 227-815-6	0 - 10
vanillin	204-465-2	0 - 10

4. First aid measures

4.1 Description of first aid measures

- General advice : Take Hazard and Precautionary phrases (section 2) into account.
- If inhaled : Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice.
- In case of skin contact : Remove contaminated clothes. Wash thoroughly with water (and soap). Contact physician if symptoms persist.

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In case of eye contact : Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist.

If swallowed : Rinse mouth with water and obtain medical advice.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms : No information available.

Risks : No information available.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : No information available.

5. Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Carbondioxide, dry chemical, foam.

Unsuitable extinguishing media : Do not use a direct waterjet on burning material.

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting : Water may be ineffective.

5.3 Advice for firefighters

Further information : Standard procedure for chemical fires.

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a major spill.

6.2 Environmental precautions

Environmental precautions : Keep away from drains, surface- and groundwater and soil.

6.3 Methods and materials for containment and cleaning up

Methods for cleaning up : Clean up spillage promptly. Remove ignition sources. Provide

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adequate ventilation. Avoid excessive inhalation of vapours. Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

6.4 Reference to other sections

Prevent spreading over a wide area (e.g. by containment or oil barriers).

7. Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use.

If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees.

Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.

Advice on protection against fire and explosion : Keep away from ignition sources and naked flame.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in a cool, dry, ventilated area away from heat sources. Keep containers upright and tightly closed when not in use.

7.3 Specific end use(s)

Specific use(s) : No information available.

8. Exposure controls/personal protection

8.1 Control parameters

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Components with workplace control parameters

Components	EC-No.	Exposure limit(s)
limonene, dipentene, d-limonene, (R)-p-mentha-1,8-diene	205-341-0, 227-813-5, 227-815-6	TWA
vanillin	204-465-2	TWA

DNEL

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one

: End Use: Workers
 Exposure routes: Skin contact
 Potential health effects: Long-term local effects
 Exposure time: 8 h
 Value: 0,1011 mg/cm²

End Use: Workers
 Exposure routes: Skin contact
 Potential health effects: Long-term systemic effects
 Exposure time: 8 h
 Value: 1,73 mg/kg bw/day

End Use: Workers
 Exposure routes: Inhalation
 Potential health effects: Long-term systemic effects

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Exposure time: 8 h
Value: 1,76 mg/m3

1,3,4,6,7,8-hexahydro-
4,6,6,7,8,8-
hexamethylindeno[5,6-c]pyran : End Use: Workers
Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 60 mg/kg bw/day

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 22 mg/m3

End Use: General population
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 6,5 mg/m3

End Use: General population
Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 36 mg/kg bw/day

End Use: General population
Exposure routes: Ingestion
Potential health effects: Long-term systemic effects

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2-ethyl-4-(2,2,3-trimethyl-3-cyclopenten-1-yl)-2-buten-1-ol : End Use: Workers
 Exposure routes: Inhalation
 Potential health effects: Long-term systemic effects
 Value: 21 mg/m³
 REACH data

End Use: Workers
 Exposure routes: Dermal
 Potential health effects: Long-term systemic effects
 Value: 6 mg/kg bw/day
 REACH data
 End Use: General population
 Exposure routes: Inhalation
 Potential health effects: Long-term systemic effects
 Value: 5,2 mg/m³
 REACH data
 End Use: General population
 Exposure routes: Dermal
 Potential health effects: Long-term systemic effects
 Value: 3 mg/kg bw/day
 REACH data
 End Use: General population
 Exposure routes: Oral
 Potential health effects: Long-term systemic effects
 Value: 3 mg/kg bw/day
 REACH data
 α-methyl-1,3-benzodioxole-5-propionaldehyde : End Use: Workers
 Exposure routes: Inhalation

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Potential health effects: Long-term systemic effects

Value: 1,2 mg/m³

REACH data

End Use: Workers

Exposure routes: Dermal

Potential health effects: Long-term systemic effects

Value: 0,17 mg/kg bw/day

REACH data

End Use: Workers

Exposure routes: Dermal

Potential health effects: Long-term local effects

Value: 0,01 mg/cm²

REACH data

End Use: General population

Exposure routes: Inhalation

Potential health effects: Long-term systemic effects

Value: 0,29 mg/m³

REACH data

End Use: General population

Exposure routes: Dermal

Potential health effects: Long-term systemic effects

Value: 0,083 mg/kg bw/day

REACH data

End Use: General population

Exposure routes: Dermal

Potential health effects: Long-term local effects

Value: 0,005 mg/cm²

REACH data

End Use: General population

Exposure routes: Oral

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coumarin

Potential health effects: Long-term systemic effects
Value: 0,17 mg/kg bw/day
REACH data
: End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Value: 6,68 mg/m3

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Acute systemic effects
No hazard identified
End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term local effects
No hazard identified
End Use: Workers
Exposure routes: Inhalation
Potential health effects: Acute local effects
No hazard identified
End Use: Workers
Exposure routes: Dermal
Potential health effects: Long-term systemic effects
Value: 0,79 mg/kg bw/day

End Use: Workers
Exposure routes: Dermal
Potential health effects: Acute systemic effects
No hazard identified
End Use: Workers
Exposure routes: Dermal

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Potential health effects: Long-term local effects

No hazard identified

End Use: Workers

Exposure routes: Dermal

Potential health effects: Acute local effects

No hazard identified

End Use: Consumers

Exposure routes: Inhalation

Potential health effects: Long-term systemic effects

Value: 1,69 mg/m³

End Use: Consumers

Exposure routes: Inhalation

Potential health effects: Acute systemic effects

No hazard identified

End Use: Consumers

Exposure routes: Inhalation

Potential health effects: Long-term local effects

No hazard identified

End Use: Consumers

Exposure routes: Inhalation

Potential health effects: Acute local effects

No hazard identified

End Use: Consumers

Exposure routes: Dermal

Potential health effects: Long-term systemic effects

Value: 0,39 mg/kg bw/day

End Use: Consumers

Exposure routes: Dermal

Potential health effects: Acute systemic effects

No hazard identified

End Use: Consumers

Exposure routes: Dermal

Potential health effects: Long-term local effects

No hazard identified

End Use: Consumers

Exposure routes: Dermal

Potential health effects: Acute local effects

No hazard identified

End Use: Consumers

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pin-2(10)-ene

Exposure routes: Oral
Potential health effects: Long-term systemic effects
Value: 0,39 mg/kg bw/day

End Use: Consumers
Exposure routes: Oral
Potential health effects: Acute systemic effects
No hazard identified, REACH data

: End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Value: 5,69 mg/m3

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Acute systemic effects
No hazard identified

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term local effects
No hazard identified

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Acute local effects
No hazard identified

End Use: Workers
Exposure routes: Dermal
Potential health effects: Long-term systemic effects
Value: 0,8 mg/kg bw/day

End Use: Workers

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Exposure routes: Dermal
Potential health effects: Acute systemic effects
No hazard identified
End Use: Workers
Exposure routes: Dermal
Potential health effects: Long-term local effects
Value: 54 µg/cm²

End Use: Workers
Exposure routes: Dermal
Potential health effects: Acute local effects
No hazard identified
End Use: Consumers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Value: 1 mg/m³

End Use: Consumers
Exposure routes: Inhalation
Potential health effects: Acute systemic effects
No hazard identified
End Use: Consumers
Exposure routes: Inhalation
Potential health effects: Long-term local effects
No hazard identified
End Use: Consumers
Exposure routes: Inhalation
Potential health effects: Acute local effects
No hazard identified
End Use: Consumers
Exposure routes: Dermal
Potential health effects: Long-term systemic effects
Value: 0,3 mg/kg bw/day

End Use: Consumers
Exposure routes: Dermal
Potential health effects: Acute systemic effects
No hazard identified
End Use: Consumers
Exposure routes: Dermal
Potential health effects: Long-term local effects

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Value: 27 µg/cm²

End Use: Consumers

Exposure routes: Dermal

Potential health effects: Acute local effects

No hazard identified

End Use: Consumers

Exposure routes: Oral

Potential health effects: Long-term systemic effects

Value: 0,3 mg/kg bw/day

End Use: Consumers

Exposure routes: Oral

Potential health effects: Acute systemic effects

No hazard identified, REACH data

cinnamaldehyde

: End Use: Workers

Exposure routes: Inhalation

Potential health effects: Long-term systemic effects

Value: 13,6 mg/m³

End Use: Workers

Exposure routes: Inhalation

Potential health effects: Acute systemic effects

No hazard identified

End Use: Workers

Exposure routes: Inhalation

Potential health effects: Long-term local effects

No hazard identified

End Use: Workers

Exposure routes: Inhalation

Potential health effects: Acute local effects

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No hazard identified
End Use: Workers
Exposure routes: Dermal
Potential health effects: Long-term systemic effects
Value: 3,85 mg/kg bw/day

End Use: Workers
Exposure routes: Dermal
Potential health effects: Acute systemic effects
not required

End Use: Workers
Exposure routes: Dermal
Potential health effects: Long-term local effects
No hazard identified

End Use: Workers
Exposure routes: Dermal
Potential health effects: Acute local effects
No hazard identified
End Use: Consumer use
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Value: 2,4 mg/m³

End Use: Consumer use
Exposure routes: Inhalation
Potential health effects: Acute systemic effects
No hazard identified

End Use: Consumer use
Exposure routes: Inhalation
Potential health effects: Long-term local effects
No hazard identified

End Use: Consumer use
Exposure routes: Inhalation
Potential health effects: Acute local effects
No hazard identified

End Use: Consumer use
Exposure routes: Dermal
Potential health effects: Acute local effects
No hazard identified

End Use: Consumer use
Exposure routes: Dermal

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Potential health effects: Long-term local effects

No hazard identified

End Use: Consumer use

Exposure routes: Dermal

Potential health effects: Acute systemic effects
not required

End Use: Consumer use

Exposure routes: Dermal

Potential health effects: Long-term systemic effects

Value: 1,37 mg/kg bw/day

End Use: Consumer use

Exposure routes: Oral

Potential health effects: Long-term systemic effects

Value: 1,37 mg/kg bw/day

REACH data

PNEC

1-(1,2,3,4,5,6,7,8-Octahydro-
2,3,8,8-tetramethyl-2-
naphthyl)ethan-1-one

: Fresh water
Value: 0,0028 mg/l

Marine water
Value: 0,00028 mg/l

Fresh water sediment
Value: 3,73 mg/kg dry weight (d.w.)

Marine sediment
Value: 0,75 mg/kg dry weight (d.w.)

Soil
Value: 0,705 mg/kg dry weight (d.w.)

1,3,4,6,7,8-hexahydro-
4,6,6,7,8,8-
hexamethylindeno[5,6-c]pyran

: Fresh water
Value: 0,0044 mg/l

Marine water
Value: 0,00044 mg/l

Fresh water sediment
Value: 2 mg/kg dry weight (d.w.)

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	Marine sediment Value: 0,394 mg/kg dry weight (d.w.)
	Soil Value: 0,31 mg/kg dry weight (d.w.)
2-ethyl-4-(2,2,3-trimethyl-3-cyclopenten-1-yl)-2-buten-1-ol	: Fresh water Value: 8,8 µg/l REACH data
	Marine water Value: 0,88 µg/l REACH data
	Sewage treatment plant Value: 1 mg/l REACH data
	Fresh water sediment Value: 1,05 mg/kg dry weight (d.w.) REACH data
	Marine sediment Value: 0,105 mg/kg dry weight (d.w.) REACH data
	Soil Value: 0,206 mg/kg dry weight (d.w.) REACH data
	Secondary Poisoning Value: 20 mg/kg REACH data
α-methyl-1,3-benzodioxole-5-propionaldehyde	: Fresh water Value: 0,005 mg/l REACH data
	Marine water Value: 0,001 mg/l

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REACH data

Sewage treatment plant

Value: 10 mg/l

REACH data

Fresh water sediment

Value: 0,057 mg/kg dry weight (d.w.)

REACH data

Marine sediment

Value: 0,006 mg/kg dry weight (d.w.)

REACH data

Soil

Value: 0,008 mg/kg dry weight (d.w.)

REACH data

coumarin

: Fresh water

Value: 19 µg/l

Marine water

Value: 1,9 µg/l

Sewage treatment plant

Value: 6,4 mg/l

Fresh water sediment

Value: 0,15 mg/kg dry weight (d.w.)

Marine sediment

Value: 0,015 mg/kg dry weight (d.w.)

Soil

Value: 0,018 mg/kg dry weight (d.w.)

REACH data

pin-2(10)-ene

: Fresh water

Value: 1,004 µg/l

Marine water

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Value: 0,1 µg/l

Sewage treatment plant

Value: 3,26 mg/l

Fresh water sediment

Value: 0,337 mg/kg dry weight (d.w.)

Marine sediment

Value: 0,034 mg/kg dry weight (d.w.)

Soil

Value: 0,067 mg/kg dry weight (d.w.)

REACH data

cinnamaldehyde

: Fresh water

Value: 0,021 mg/l

REACH data

Marine water

Value: 0,002 mg/l

REACH data

8.2 Exposure controls

Engineering measures

Where appropriate, use closed systems to transfer and process this material.

If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

Personal protective equipment

Respiratory protection

: Use local exhaust ventilation around open tanks and other open sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures. No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

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If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

- a) while engineering controls and appropriate safe work practices and/or procedures are being implemented; or
- b) during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or
- c) if normal operational workplace vapor concentration in the air is increased due to heat ;
- d) during emergencies; or
- e) if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.

- Hand protection : Avoid skin contact. Use chemically resistant gloves.
- Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.
- Hygiene measures : To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material.
To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize exposure.

Environmental exposure controls

- General advice : Keep away from drains, surface- and groundwater and soil.

9. Physical and chemical properties

9.1 Information on basic physical and chemical properties

- Appearance : liquid
- Odour : conforms to standard
- Odour Threshold : not determined

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Flash point	: 83 °C
Lower explosion limit	: not determined
Upper explosion limit	: not determined
Flammability (solid, gas)	: not determined
Oxidizing properties	: not determined
Auto-ignition temperature	: not determined
pH	: not determined
Melting point	: not determined
Boiling point	: not determined
Vapour pressure	: 0,21 hPa Calculated
Density	: not determined
Water solubility	: not determined
Partition coefficient: n-octanol/water	: not determined
Solubility in other solvents	: not determined
Viscosity, dynamic	: not determined
Viscosity, kinematic	: not determined
Relative vapour density	: not determined
Evaporation rate	: not determined

9.2 Other information

Refractive index : not determined

10. Stability and reactivity

10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Note: Presents no significant reactivity hazard, by itself or in contact with water. Avoid contact with strong acids, alkali or oxidizing agents.

10.4 Conditions to avoid

Conditions to avoid : Direct sources of heat.

10.5 Incompatible materials

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Materials to avoid : Avoid contact with strong acids, alkali or oxidizing agents.

10.6 Hazardous decomposition products

Hazardous decomposition products : Carbon monoxide and unidentified organic compounds may be formed during combustion.

11. Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Skin corrosion/irritation

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one : Species: human
Result: Skin irritation
Method: OECD 439

Serious eye damage/eye irritation

1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one : Species: Chicken eye
Result: Eye irritation
Method: OECD 438

Respiratory or skin sensitisation

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one : LLNA
Species: Mouse
Result: Causes sensitisation.
Method: OECD 429

Germ cell mutagenicity

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Carcinogenicity

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Reproductive toxicity

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Information on ingredients with chronic toxicity hazard(s) is available upon request.

Target Organ Systemic Toxicant - Single exposure

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Target Organ Systemic Toxicant - Repeated exposure

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Aspiration hazard

No test data available. Ingredients with an aspiration hazard are identified in section 3.

12. Ecological information

12.1 Toxicity

M-Factor

[3R-(3.alpha.,3a,7,8a.alpha.)]- : 1
1-(2,3,4,7,8,8a-hexahydro-
3,6,8,8-tetramethyl-1H-3a,7-
methanoazulen-5-yl)ethan-1-one

12.2 Persistence and degradability

Biodegradability

[3R-(3 α ,3a,7,8 α)]-1- : Result: Not readily biodegradable.
(2,3,4,7,8,8a-hexahydro-3,6,8,8- 36 %
tetramethyl-1H-3a,7- Method: Directive 67/548/EEC Annex V, C.4.C.
methanoazulen-5-yl)ethan-1-
one

12.3 Bioaccumulative potential

12.4 Mobility in soil

Mobility : No data available

12.5 Results of PBT and vPvB assessment

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This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects

Additional ecological information : There is no data available for this product.

13. Disposal considerations

13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

ADR

UN number : 3082
Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Octahydro Tetramethyl Naphthalenyl Ethanone)
Labels : 9
Packing group : III
Environmentally hazardous : yes

IATA

UN number : 3082
Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Octahydro Tetramethyl Naphthalenyl Ethanone)
Labels : 9
Packing group : III
Environmentally hazardous : yes

IMDG

UN number : 3082
Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Octahydro Tetramethyl Naphthalenyl Ethanone)
Labels : 9
Packing group : III
Marine pollutant : yes (Octahydro Tetramethyl Naphthalenyl Ethanone)

Special precautions for : No special precautions required.

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user

15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Water contaminating class : (Germany)	WGK 3highly hazardous to water
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15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

16. Other information

Full text of H-Statements referred to under sections 2 and 3.

H226	Flammable liquid and vapour.
H302	Harmful if swallowed.
H304	May be fatal if swallowed and enters airways.
H312	Harmful in contact with skin.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.
H361	Suspected of damaging fertility or the unborn child.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.
H413	May cause long lasting harmful effects to aquatic life.

Further information

In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110]. In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace". Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.

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The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration $\geq 1.0\%$ due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

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Czech Republic	+420 224 919 293 / +420 224 915 402
Denmark	+45 82 12 12 12
Estonia	16662 (National), International (+372) 626 93 90
Finland	+358 9 471977
France	+ 33 (0)1 45 42 59 59
Germany	+31 13 4642 211
Greece	+31 13 4642 211
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Poland	+31 13 4642 211
Portugal	808 250 250

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Sweden	+46 112
United Kingdom	+44 111 (England, Wales & Scotland)